CDA Spring 2026

Line Clemmensen

March 10, 2026

Week 6: Ensemble models

1 Derive the variance for an average of B i.i.d. random variables, each with variance σ^2. What happens when B goes to infinity, i.e. we increase the number of trees?

(a) Tip: $Var(X+Y) = Var(X) + Var(Y)$ if X and Y are independent.

(b) Tip: $Var(aX) = a^2 Var(X)$ for a constant a.

2 Derive the variance for an average of B i.d. (not independent) random variables, each with variance σ^2, and positive pairwise correlation ρ. What happens when B goes to infinity, i.e. we increase the number of trees?

(a) Tip: $Var(X+Y) = COV(X+Y, X+Y) = COV(X,X) + COV(X,Y) + COV(Y,X) + COV(Y,Y)$.

(b) Tip: correlation $= \rho$ => covariance $= \rho\sigma^2$.

3 How does this relate to the following plot from ESL?

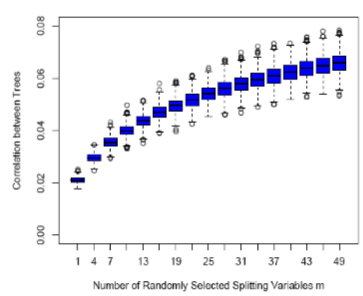

FIGURE 15.9. *Correlations between pairs of trees drawn by a random-forest regression algorithm, as a function of m. The boxplots represent the correlations at 600 randomly chosen prediction points x.*

4 Load the zip data and make a classification using random forests. As a minimum you need to tune the parameters: The number of trees, the number of variables to sample, and the tree size. Make an effort of explaining what happens when you tune each of the parameters. Which one is more important? Additionally answer:

(a) How should you set the parameters to run bagging?

(b) How should you set the parameters to run CART?

(c) Plot variable importances and a variable correlation matrix. How do you interpret these?

(d) Add five randomly generated, white noise, variables to the dataset and investigate how you can use their information to derive a cut-off value to select varaibles based on the variable importances.

5 Fit boosting on the zip data. You should tune the learning rate, the number of models, and also the individual models to obtain the best classification rates. Which tuning parameters are the most important for obtaining good performance with boosting?